6.2 Special Right Triangles

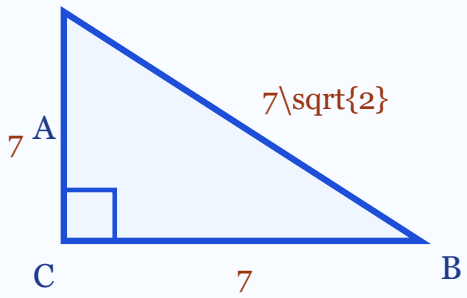
Two right-triangle shapes recur so often that their side ratios are worth memorizing. A 45-45-90 **triangle** is an isosceles right triangle: if each leg is x, the Pythagorean Theorem gives hypotenuse $= x\sqrt{2}$. A 30-60-90 **triangle** comes from bisecting an equilateral triangle: if the equilateral side is $2x$, the altitude splits it into two right triangles with short leg x (opposite 30°), long leg $x\sqrt{3}$ (opposite 60°), and hypotenuse $2x$ (the original side). Once you recognize either pattern from a single given length, every other side follows immediately — no need to re-derive the Pythagorean Theorem each time. This worksheet derives both patterns and then applies them to triangles, squares, and equilateral triangles.

Examples

Example 1: 45-45-90 — find the hypotenuse

A 45-45-90 triangle has legs of length 7. Find the hypotenuse.

Solution



Both legs equal; hypotenuse is leg $\times\sqrt{2}$.

Identify the pattern: legs $x = 7$; hypotenuse $= x\sqrt{2}$.

Substitute: hypotenuse $= 7\sqrt{2}$.

Check with the Pythagorean Theorem: $7^2 + 7^2 = 49 + 49 = 98$, and $(7\sqrt{2})^2 = 49 \times 2 = 98$ ✓.

Answer: $7\sqrt{2}$.

Example 2: 45-45-90 — find a leg

A 45-45-90 triangle has hypotenuse $10\sqrt{2}$. Find each leg.

Solution

Identify the pattern: hypotenuse $= x\sqrt{2} = 10\sqrt{2}$, so $x = 10$.

Each leg $= 10$.

Check: $10^2 + 10^2 = 200$, and $(10\sqrt{2})^2 = 200$ ✓.

Answer: each leg $= 10$.

Example 3: Derive the 45-45-90 pattern

Prove that an isosceles right triangle with legs x has hypotenuse $x\sqrt{2}$.

Solution

Let the two legs both equal x (isosceles right triangle, right angle between them).

Apply the Pythagorean Theorem: $c^2 = x^2 + x^2 = 2x^2$.

Solve for c: $c = \sqrt{2x^2} = x\sqrt{2}$ (length is positive).

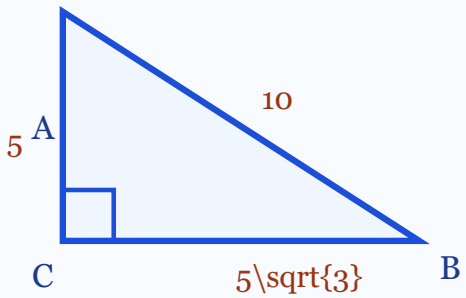
Conclusion: the hypotenuse of any 45-45-90 triangle is always leg $\times\sqrt{2}$.

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Example 4: 30-60-90 from the short leg

A 30-60-90 triangle has short leg (opposite 30°) of length 5. Find the long leg and hypotenuse.

Solution



Short leg x, long leg $x\sqrt{3}$, hypotenuse $2x$.

Identify the pattern: short leg $= x = 5$.
Long leg $= x\sqrt{3} = 5\sqrt{3}$; hypotenuse $= 2x = 10$.
Check: $5^2 + (5\sqrt{3})^2 = 25 + 75 = 100 = 10^2$ ✓.
Answer: long leg $= 5\sqrt{3}$; hypotenuse $= 10$.

Example 5: 30-60-90 from the hypotenuse

A 30-60-90 triangle has hypotenuse 14. Find both legs.

Solution

Identify the pattern: hypotenuse $= 2x = 14$, so $x = 7$.
Short leg $= x = 7$; long leg $= x\sqrt{3} = 7\sqrt{3}$.
Check: $7^2 + (7\sqrt{3})^2 = 49 + 147 = 196 = 14^2$ ✓.
Answer: short leg $= 7$; long leg $= 7\sqrt{3}$.

Example 6: 30-60-90 from the long leg

A 30-60-90 triangle has long leg (opposite 60°) of length $6\sqrt{3}$. Find the short leg and hypotenuse.

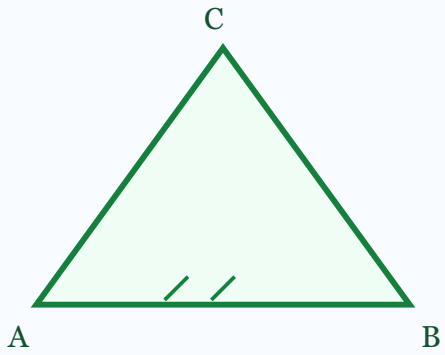
Solution

Identify the pattern: long leg $= x\sqrt{3} = 6\sqrt{3}$, so $x = 6$.
Short leg $= x = 6$; hypotenuse $= 2x = 12$.
Check: $6^2 + (6\sqrt{3})^2 = 36 + 108 = 144 = 12^2$ ✓.
Answer: short leg $= 6$; hypotenuse $= 12$.

Example 7: Derive the 30-60-90 pattern

Prove that bisecting an equilateral triangle with side $2x$ produces a 30-60-90 triangle with legs x and $x\sqrt{3}$.

Solution



The altitude of an equilateral triangle bisects the base and the apex angle.

In equilateral $\triangle ABC$ with side $2x$, drop the altitude from A to the midpoint D of $\overline{BC}$.
Since the altitude of an equilateral triangle bisects the base and the apex angle: $BD = x$, and $\angle BAD = 30°$, $\angle ABD = 60°$, $\angle ADB = 90°$.
Apply the Pythagorean Theorem in $\triangle ABD$: $AD^2 = AB^2 - BD^2 = (2x)^2 - x^2 = 4x^2 - x^2 = 3x^2$
.
$AD = x\sqrt{3}$.
Conclusion: $\triangle ABD$ has short leg x (opposite 30°), long leg $x\sqrt{3}$ (opposite 60°), hypotenuse $2x$ — the 30-60-90 pattern.

Example 8: Square diagonal

A square has side length 4. Find the length of the diagonal.

Solution

A diagonal splits the square into two 45-45-90 triangles with legs 4.

Diagonal $=$ hypotenuse $= x\sqrt{2} = 4\sqrt{2}$.

Answer: $4\sqrt{2}$.

Example 9: Equilateral triangle altitude and area

An equilateral triangle has side length 10. Find the altitude, then the area.

Solution

The altitude splits the equilateral triangle into two 30-60-90 triangles with hypotenuse 10, so $x = 5$.

Altitude $= x\sqrt{3} = 5\sqrt{3}$.

Find the area using base $= 10$ and height $= 5\sqrt{3}$: $A = \frac{1}{2}(10)(5\sqrt{3}) = 25\sqrt{3}$.

Answer: altitude $= 5\sqrt{3}$; area $= 25\sqrt{3}$ square units.

Example 10: Rectangle diagonal at 30°

A rectangle's diagonal is 20 cm and makes a 30° angle with the longer side. Find both side lengths.

Solution

The diagonal, the longer side, and the shorter side form a 30-60-90 triangle, with the diagonal as the hypotenuse (20).

The side adjacent to the 30° angle is the longer side (opposite the 60° angle): longer side $= 20 \cdot \frac{\sqrt{3}}{2} = 10\sqrt{3}$.

The side opposite the 30° angle is the shorter side: shorter side $= 20 \cdot \frac{1}{2} = 10$.

Answer: sides are 10 cm and $10\sqrt{3}$ cm.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: 45-45-90

Legs = 9. Find the hypotenuse.

Answer: **Verification:**

Problem 2: 45-45-90

Hypotenuse = $6\sqrt{2}$. Find each leg.

Answer: **Verification:**

Problem 3: 30-60-90

Short leg = 4. Find the long leg and hypotenuse.

Answer: **Verification:**

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Problem 4: 30-60-90

Hypotenuse $= 18$. Find both legs.

Answer: **Verification:**

Problem 5: Square

Square side 5. Find the diagonal.

Answer: **Verification:**

Problem 6: 30-60-90

Long leg $= 5\sqrt{3}$. Find the short leg and hypotenuse.

Answer: **Verification:**

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Problem 7: Equilateral

Side 8. Find the altitude.

Answer: **Verification:**

Problem 8: 45-45-90

Hypotenuse $= 14$. Find each leg.

Answer: **Verification:**

Problem 9: 30-60-90

Short leg $= \sqrt{2}$. Find the hypotenuse.

Answer: **Verification:**

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Problem 10: 45-45-90

Leg = $2\sqrt{5}$. Find the hypotenuse.

Answer: **Verification:**

Problem 11: 30-60-90

Hypotenuse = $2\sqrt{3}$. Find the short and long legs.

Answer: **Verification:**

Problem 12: Square

Diagonal = $10\sqrt{2}$. Find the side length.

Answer: **Verification:**

Problem 13: Challenge — equilateral area

Equilateral triangle side = 12. Find the altitude and the area.

Answer: **Verification:**

Problem 14: Challenge — rectangle diagonal

A rectangle's diagonal is 16 cm and makes a 60° angle with the shorter side. Find both side lengths.

Answer: **Verification:**

Problem 15: Challenge — combined patterns

A 30-60-90 triangle has short leg 6. Its long leg is the hypotenuse of a 45-45-90 triangle. Find the legs of that 45-45-90 triangle.

Answer: **Verification:**

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✓ Answer Key

1. $9\sqrt{2}$
2. 6
3. $4\sqrt{3}$; 8
4. 9; $9\sqrt{3}$
5. $5\sqrt{2}$
6. 5; 10
7. $4\sqrt{3}$
8. $7\sqrt{2}$
9. $2\sqrt{2}$
10. $2\sqrt{10}$
11. $\sqrt{3}$; 3
12. 10
13. Altitude = $6\sqrt{3}$; area = $36\sqrt{3}$
14. 8 cm and $8\sqrt{3}$ cm
15. $3\sqrt{6}$ each

Great work — keep practising!